

Program : Diploma in Architecture	
Course Code : 4183	Course Title: Building Services
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To define and identify engineering service systems in a building for providing comfort and protection to the residents.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Services in buildings		Building construction and Materials	3
Detailing of services		Working Drawing I Masonry and service Lab	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Identify suitable systems for electrification in buildings	15	Understanding
CO2	Identify and select suitable systems for water supply and sanitation in a building	17	Understanding
CO3	Identify suitable systems for Ventilation, Air Conditioning and acoustics in buildings	14	Understanding
CO4	Identify suitable systems for firefighting and vertical transportation in buildings	12	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3				3		
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify suitable systems for electrification in buildings		
M1.01	Classify wiring systems, wires, lighting accessories and fittings used in electrification systems	9	Understanding
M1.02	Familiarize with different methods to ensure safety in electrification of buildings	3	Understanding
M1.03	Explain wiring diagrams for residential building	3	Understanding

Contents:

Building services - Introduction - necessity - Electrical wiring - introduction - wiring system - consumer's terminal - property of supplier - systems of wiring - cleat wiring- wooden casing and capping wiring - CTS or PVC or TRS wiring - lead sheathed or metal sheathed wiring - conduit wiring surface or open - recessed or concealed - wires and cables - difference between wires and cables - parts of a cable - conductor - insulation - protective covering - conductor materials used in cables - copper, Aluminium insulating materials - properties - types - rubber - VIR - impregnated paper - PVC - silk and cotton - types of wires and cables used in internal wiring - according to conductor used - copper conductor cables - Aluminium conductor cables - classification of cables and wires according to number of cores - single core cable - twin core cable - three core cable - classification of cables and wires according to voltage grading - 250/440 v650/1100v - definition and positioning of equipment like meter board, main switch board, distribution board, Lighting accessories and fittings - switches - one way - two way - ceiling rose - socket outlet - plugs - lamp holder - Electrical shock - causes - effects of electrical current on human body precautions - protective devices - fuses - MCB - ELCB - RCCB - earthing - purpose - specification regarding earthing - factors affecting earth resistance - methods of earthing- strip-rod-pipe-plate earthing - lightning - different methods of lightning protection

Conventional symbols used in electrical installations for residential buildings - wiring diagram for small residential building.

CO2	Identify suitable systems for water supply and sanitation in a building		
M2.01	Classify pipes, pipe appurtenances and fittings used in water supply system.	2	Understanding
M2.02	Familiarize with hot water supply and rain water harvesting in a residential building	2	Understanding
M2.03	Familiarize with components of service connection in a residential building	2	Understanding
M2.04	Explain house drainage systems	3	Understanding
M2.05	Identify sanitary systems in a building	4	Understanding
M2.06	Develop house drainage plan and service plan of a building	4	Applying
	Series Test - I	1	

Contents:

Water supply system - types of pipes - cast iron pipes, steel pipes, R.C.C pipes, pre - stressed concrete pipes, AC pipes, copper, brass, lead, wrought iron, PVC, Hume steel pipes, polyethylene pipes - pipe appurtenances - definition valves - types - stop valve, check valve, air relief valve, pressure relief valve, scour valve, fire hydrant, plug valves, water meter - pipe fittings - objective bow, tee, reducer, lateral, cross, cap, coupling, union, nipple specifications - Layout of water supply system in a house-hot and cold water supply system in a building - water heater - objectives - types - Service connections in a building - storage tanks - function-types-underground and overhead tanks- rain water harvesting system

Sanitary engineering-definition-terminology-house drainage-parts of a drainage system-traps-function- requirements of a good trap-classification of traps according to shape-S, P and Q- classification of traps according to uses-floor trap, gully trap, intercepting trap-grease and oil trap- Systems of plumbing-single stack partially ventilated, two pipe, one pipe, single stack-sanitary fixtures-water closet, urinal, sink, wash basin, bath tubs, shower cubicles, flushing cisterns-manholes-function-location-construction-septic tank-working-design and construction details-soak pit-function-construction details-layout of house drainage- Service plan of a building

CO3	Identify suitable systems for Ventilation and Air Conditioning in buildings		
M3.01	Explain systems of ventilation	4	Understanding
M3.02	Outline mechanical ventilation systems	2	Understanding
M3.03	Explain purpose of air conditioning and familiarize commonly used A/C systems	4	Understanding
M3.04	Familiarize essential components of an air-conditioner	4	Understanding

Contents:

Ventilation - necessity - functional requirements - systems of ventilation natural - wind effect and stack effect - mechanical ventilation systems - exhaust, plenum, balanced - Air

conditioning - purpose of air conditioning-systems of air conditioning - central, self - contained, semi contained, combined - essentials of an air - conditioned system - filter, heating, cooling, humidification, dehumidification, air circulation			
CO4	Identify suitable systems for fire protection, vertical transportation, and Acoustics for buildings		
M4.01	Select suitable fire-fighting measures for a building	3	Understanding
M4.02	Identify components of lifts and escalators and familiarize their working	3	Understanding
M4.03	Outline acoustical requirements of a building.	3	Understanding
M4.04	Familiarize with acoustical defects and solutions	3	Understanding
	Series Test - II	1	
Contents: Firefighting services - classification of fire - class A, class B, class C, class D, class E - Classification of buildings according to fire load - causes and effects of fire-important considerations in fire protection - limiting fire spread - fire resistant construction-fire protection systems - alarm, extinguishers, escape routes- Lifts and escalators - lifts - types of lifts - traction, hydraulic - passenger lifts, bed lifts, dumb waiters - structural provision for lift installation - lift pit, lift well, machine room, hoist way - arrangement of lifts - escalators parts - truss, landing platform, steps, tracks - arrangement - parallel, cross - cross, multiple parallel- Acoustics - acoustics of buildings - characteristics of audible sound - frequency, loudness, tone - behavior of sound transmission, reflection, absorption - acoustical defects, causes and remedies - echo, reverberation, sound foci and dead spots, insufficient loudness, noise nuisance-Sabine's formula - general principles and factors in acoustical design - site selection and planning, volume, shape, treatment of interior surfaces, reverberation, seat, seating arrangement and audience, sound absorption - acoustical materials - porous absorbents, resonant panel absorbent, cavity resonators, composite type resonators - recommendations for good acoustical design of buildings like open air theatre, cinema theatre, radio broad casting stations, concert halls, class lecture rooms, public lecture halls and multi - purpose theatres			

Text / Reference:

T/R	Book Title/Author
T1	Water supply and Sanitary Engineering- Rangwala
T2	Building construction - Arora and Bindra
T3	Building construction -Varghese
T4	Building construction-Jha